HRISHIKESH BELAGALI

Undergraduate Research Assistant

East Lansing | github.com/hkbelagali | belagal1@msu.edu | /in/hkbel

SUMMARY

PhD applicant in Computer Science with interests in quantum algorithms, molecular dynamics, computational physics, and machine learning.

SKILLS

Languages: Python, Rust, C++, LaTeX

Packages: NumPy, PyTorch, JAX, scipy, Qiskit, Cirq, ruff, mypy, pytest

Softwares: LAMMPS, OVITO, NAMD, VMD

EDUCATION

Computer Science (B.S) — **Michigan State University Honors College**
Expected Graduation: 2027
Current GPA: 4.0

RELEVANT COURSEWORK

- Graduate Quantum Algorithms
- Graduate Computational Foundations of Artificial Intelligence
- Graduate Tensor Networks (Scheduled)
- Linear Algebra

EXPERIENCE

2026– **Undergraduate Research Assistant** — **MSU-Q**

- Conducted research on fermionic backpropagation, a method for performing hybrid Schrödinger-Heisenberg simulation of large-scale unitary cluster Jastrow circuits.
- Implemented fermionic backpropagation into *ffsim*, a Python package for simulating fermionic quantum systems.
- Primary developer of *propaq*, a Python package for Heisenberg propagation, designed for large-scale simulation on HPC clusters.

2025–2026 **Honors Research Scholar** — **MSU Honors College/MSU-Q**

- Conducted research on algorithms for finding optimal compilation subgroups under the guidance of Professor Ryan LaRose.
- Investigated computational group theory methods for membership in unitary subgroups.
- Implemented compilation optimization for generalized subgroups using Python and Cirq.

2024– **Undergraduate Research Assistant** — **Vermaas Lab, Plant Research Laboratory**

- Simulated the effects of small molecules like butanol on *Clostridium acetobutylicum* and *Zymomonas mobilis* membrane dynamics.
- Investigated the interaction of the chloroplast targeting protein (At4g16060) with thylakoid membranes using molecular dynamics.
- Analyzed the effect of PDLP-5 (Plasmodesmata Located Protein) on water flow rates through PIP1;4 aquaporin tetramer using grid-steered molecular dynamics.

PUBLICATIONS

[1] Hrishikesh Belagali, Ryan LaRose. "propaq – A Python package for Heisenberg propagation", manuscript in preparation.

[2] Hrishikesh Belagali, Thomas Van Camp, R. Pradeep, Sourin Das, Namit Anand, Ryan LaRose, "Efficient classical simulation of large-scale unitary cluster Jastrow circuits," *arXiv*, 2026, manuscript submitted to *Physical Review Letters* (pending review). Available: https://arxiv.org/abs/2607.21337

[3] Hrishikesh Belagali, Nitin Kumar Singh, Josh V. Vermaas, "The Nanoscale Impact of Cyclopropane Fatty Acids and Hopanoids on Alcohol Tolerance in *Clostridium acetobutylicum* and *Zymomonas mobilis*," *ChemRxiv*, 2026, manuscript submitted to *ACS JPCB* (pending review). Available: https://chemrxiv.org/doi/full/10.26434/chemrxiv.15004463/v1

CONFERENCES

2026
ACS, IL

Molecular mechanisms of alcohol tolerance in Clostridium acetobutylicum and Zymomonas mobilis: The role of cyclopropane fatty acids and hopanoids — **acs.digitellinc.com**
Poster presented at ACS Fall 2026 Conference

2026 *MQC, IN*	**Efficient classical simulation of large-scale unitary cluster Jastrow circuits** Poster presented at MQC Entanglement 2026
2026 *GLBRC, WI*	**The Nanoscale Impact of Cyclopropane Fatty Acids and Hopanoids on Alcohol Tolerance in *Clostridium acetobutylicum* and *Zymomonas mobilis*** Poster presented at Great Lakes Bioenergy Research Center Annual Science Meeting.
2026 *MSU UURAF*	**Reducing T Gate Counts via Gateset Rotation** Poster presented at MSU Undergraduate Research and Arts Forum. Awarded first place in the Computational Mathematics, Science and Engineering division.

OPEN SOURCE CONTRIBUTIONS

propaq
Python, Rust

propaq **github.com**

Primary developer of *propaq*, a Python package for Heisenberg propagation, designed for large-scale simulation on HPC clusters.

- Implmented thread-safe kernels and compact representation of Pauli and Majorana strings in Rust for optimal multi-threaded simulation.
- Developed a Python interface integrating *propaq* with Qiskit and Cirq for seamless simulation workflows using pyo3 bindings.
- Designed a plugin system for developing custom noise and truncation policies in Python, Rust, C, and Julia to enable user customization with minimal overhead.

ffsim
Python, Rust

ffsim **github.com**

Contributed to *ffsim*, a Python package for simulating fermionic quantum systems.

- Implemented fermionic backpropagation, from "Efficient classical simulation of large-scale unitary cluster Jastrow circuits" (arXiv:2607.21337) into *ffsim* for fast unitary cluster Jastrow ansatz simulation.
- Created a JAX-based implementation for GPU-accelerated variational optimization.

QuTiP
Python

Quantum Toolbox in Python (QuTiP) **github.com**

Contributed to QuTiP, an open-source software for simulating quantum systems.

- Implemented functionality to return eigenkets as an operator in the Qobj class.

Python
NumPy

aqua-blue **github.com/pypi.org**

Secondary maintainer of aqua-blue, a package building echo state networks.

- Implemented reservoir state time-lag, sparsity and spectral radius to improve prediction accuracy.
- Introduced timezone-aware datetime support for seamless interfacing with data pipelines.
- Demonstrated prediction capability by predicting quantum time evolution of Gaussian wave packets.

Python
NumPy
hyperopt

aqua-blue-hyperopt **github.com/pypi.org**

Primary maintainer of aqua-blue-hyperopt, a package implementing hyperparameter optimization functionality for aqua-blue models.

- Created modular hyperparameter optimization functionality for aqua-blue reservoir computers.
- Implemented custom loss function capabilities for flexible model evaluation.

tce-lib
Python

tce-lib **github.com**

Contributed to tce-lib, a Python package for building tensor cluster expansion models for concentrated alloys.

- Implemented lazy evaluation of Monte Carlo trajectories for efficient simulation on HPC clusters.

HONORS AND AWARDS

- Wielenga Research Scholarship, Honors College (2025-2026)
- Dean's List (Fall 2024, Spring 2025, Fall 2025, Spring 2026)
- Senator Joe and Mary Conroy Endowed Scholarship, College of Engineering (2024)

PROJECTS

Python
Qiskit

Adiabatic Quantum Computation **github.com**

Implemented a quantum adiabatic algorithm to solve QUBO type problems using the Trotter-Suzuki approximation and quantum Ising model formulation techniques.

Python
OVITO
pyMOL

Genetic Annealing to Determine Protein Structures **github.com**

Monte Carlo Genetic Algorithm to determine protein structures through the optimization of Irbäck's off-lattice model energy equation (RMSD < 3.0).